

Evidence appendix

All results come from one discrete-event simulator that implements the brief's network exactly (moving-receiver light time, solar exclusion on every launch, 19 links, FIFO serialization, relay processing, hop and endpoint timers, handshakes, quotas, maintenance, incidents, per-launch loss) and our v3 ledgers. Every traced run checks conservation, non-negativity, hold arithmetic and registry custody after every ledger event. A trace is a **conditional, no-loss** run unless labelled "lossy". Times are hours after the scenario start; the epoch is 2026-09-22 00:00 TDB. "Bound" and "exact" are stated wherever a probability appears.

E1. Orbital calculations

Method. Each body follows the fixed Kepler ellipse in `data.zip`: mean anomaly $M = M_0 + n \cdot t$ (n from the supplied mean motion), Kepler's equation solved by Newton iteration until the residual is below 10^{-15} rad, then rotation by ω , i , Ω into the Sun-centred J2000 ecliptic frame. Units: AU and seconds of TDB since the epoch (hours in tables). The relays use the same code with $e = 0$ and the supplied phases. **Light time:** a packet emitted at t_e from P reaches Q at the t_a solving $t_a = t_e + 499.02 \cdot |Q(t_a) - P(t_e)|$ (8.317 min/AU), found by fixed-point iteration; for Earth→A the successive corrections are 2.1e-02, 4.1e-07, 7.7e-12 s (each step shrinks by $v/c \approx 10^{-4}$), so two iterations reach 1 ms and we always iterate to 10^{-6} s. **Solar exclusion:** distance from the Sun to the segment $P(t_e) \rightarrow Q(t_a)$, blocked below 0.10 AU. Every launch, ACK and receipt is evaluated at its own emission time. Maximum epoch error against the brief's table: 7.3e-07 AU (table rounding).

Body	x (AU)	y (AU)	z (AU)	error
Mercury	-0.213426	-0.410281	-0.013955	1.8e-07
Venus	+0.679067	-0.257950	-0.042725	3.4e-07
Earth	+1.003581	-0.023693	-0.000003	5.3e-07
Mars	+0.247476	+1.526869	+0.025929	7.3e-07
Ceres	+0.375786	+2.653941	+0.014791	5.6e-07
Jupiter	-3.438179	+4.038309	+0.060149	5.2e-07
Saturn	+9.271221	+1.717485	-0.398962	3.8e-07
Uranus	+8.962475	+17.253919	-0.052129	6.0e-07
Neptune	+29.839098	+1.351785	-0.715470	3.3e-07
Relay A	+2.000000	+2.000000	+0.000000	4.4e-16
Relay B	-2.000000	+2.000000	+0.000000	4.4e-16

Worked later positions (same propagator): hour 300 and one Julian year.

Body	hour 300 (x, y, z) AU	hour 8,766 (x, y, z) AU
Earth	(+0.9822, +0.1894, -0.0000)	(+1.0036, -0.0192, -0.0000)
Mars	(+0.0802, +1.5602, +0.0307)	(-0.4854, -1.4173, -0.0178)
Ceres	(+0.2446, +2.6592, +0.0391)	(-2.4828, +0.3753, +0.4693)
Relay A	(+1.9076, +2.0883, +0.0000)	(-1.4432, +2.4326, +0.0000)
Relay B	(-2.0883, +1.9076, +0.0000)	(-2.4326, -1.4432, -0.0000)
Neptune	(+29.8370, +1.3912, -0.7162)	(+29.7556, +2.5017, -0.7374)

Receiver motion changes arrival. Rows 1–4 are the S1a funding path forward (Earth→A→Mars) and the return path for the shares (Mars→A→Earth) at their actual emission times; each relay forward departs 2 s after arrival (1 s processing + 1 s serialization). "Frozen" uses both positions at emission. The forward and return legs are not mirror images: Earth→A gains +21.4 ms because Relay A moves away during flight, while the return legs differ again.

Path	t_e (s)	flight (s)	frozen (s)	Δ ms	d (AU)	clear (AU)
Earth → A	1.000	1125.660748	1125.639333	+21.415	2.2557	1.004
A → Mars	1128.661	906.054086	905.991624	+62.462	1.8157	1.547
Mars → A	0.000	905.920379	905.947047	-26.669	1.8154	1.547
A → Earth	907.920	1125.475560	1125.575802	-100.241	2.2554	1.004
Earth → Mars	2.000	860.995877	860.951580	+44.297	1.7254	0.892
Mars → Earth	2.000	860.875291	860.951580	-76.289	1.7251	0.892
Neptune → A	1.000	13901.212047	13900.618008	+594.039	27.8570	2.828
A → Neptune	1.000	13900.598751	13900.618008	-19.257	27.8558	2.828

Timers used. T0 is the empty-queue route time; R_e = 2·T0 + 24 h; R_h = 2·flight + 60 min per hop.

Route	at h	T0 (min)	R_e (h)	R_h per hop (min)
Ea-A-Ma	0.0000	33.91	25.130	97.5, 90.2
Ma-A-Ea	1.6966	33.90	25.130	90.2, 97.5
Ce-A-Ma	0.0000	29.71	24.990	89.1, 90.2
Ne-A-Ma	0.0000	246.85	32.228	523.4, 90.2

S1. Scenario traces (no loss, epoch, maintenance on)

Traces omit only the "receipt known" bookkeeping rows (no balance changes) and Bob's publication of each observation (the next row shows Mars recording it). Each scenario is a separate run from the opening book. Completion = every intended result spendable at its owner's home. Packets count every backbone launch (data, handshake, hop receipts, ACKs, retries) plus every direct copy. Encumbered = order holds + contract margin + value in transit, per asset as peak / asset-hours. Utilisation = peak encumbered cash ÷ \$500,000. Capital efficiency = peak encumbered \$ ÷ value settled.

Scenario	Done (h)	Packets bb + dir	Tx	Pkts/tx	Peak enc. / asset-hours	Util.	Cap. eff.
S1a Earth → Mars Ares purchase	3.07	44 + 3 = 47	2	23.5	\$ 100,000 / 169,629; Ares 1,000 / 3,067	20%	1.00
S1b Earth → Ceres Belt purchase	3.16	44 + 3 = 47	2	23.5	\$ 100,000 / 166,938; Belt 1,000 / 3,164	20%	1.00
S1c Uranus → Earth Terra purchase	15.82	44 + 8 = 52	2	26.0	\$ 40,000 / 316,523; Terra 400 / 6,330	8%	1.00
S1d Futures, rising	328.17	112 + 8 = 120	3	40.0	\$ 160,000 / 52,386,684	32%	0.80
S1e Futures, falling	328.17	112 + 8 = 120	3	40.0	\$ 160,000 / 52,372,538	32%	0.80
S1f Futures Q = 2,500 (boundary)	328.17	112 + 8 = 120	3	40.0	\$ 200,000 / 65,483,355	40%	0.80
S1g Futures Q = 2,501 (binds)	—	44 + 7 = 51	2	25.5	\$ 100,040 / 16,782,137	20%	—
S1h Futures Q = 4,000 (binds)	—	0 + 0 = 0	0	—	—	0%	—
S1i \$10k repeat trading	51.06	44 + 15 = 59	7	8.4	\$ 10,000 / 105,985; Ares 15 / 51	2%	—

Scale and binding constraint. The futures hold \$160,000 = 32% of all cash in escrow for 324 h (S1d/S1e); the boundary Q = 2,500 holds \$200,000 = 40%. Q = 2,501 needs \$100,040 per side: Cara's Ceres exchange rejects her funding transfer at once (she owns \$100,000), Alice's \$100,040 is held at Mars until the 168 h deadline, released, and returned home by her withdrawal at 169.79 h. Q = 4,000 (\$160,000 each) is rejected at both homes before anything moves. Every audit passed: 91 checks across these runs.

S1a full trace: Alice (Earth) buys 1,000 Ares from Bob (Mars) at \$100

h	Actor	Knows	Does	Packet / transmission	Arrives	Financial state after
0.0003	Bob	Own inventory at the market	Sell 1,000 Ares limit \$100 GTC	Local access at Mars	0.0006	Alice \$150k at Earth Bob \$50k at Mars, 3,000 Ares home
0.0003	Mars exchange	Its own available balances	Admits Bob EQ-sell; holds 1,000 Ares	—	—	Alice \$150k at Earth Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Alice	Own Earth balance	Funds own Mars account: \$100k	Local access at Earth	0.0006	Alice \$150k at Earth Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Earth exchange	Instruction or fill; its own ledger	Debits Alice \$100k; TRANSFER EA-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ea → A h 1.1317, A → Ma h 1.4448	1.6966	Alice \$50k at Earth, \$100k in transit Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
1.6966	Mars exchange	Authentic new transfer EA-X1	Imports once; credits Alice \$100k (spendable now)	RECEIVED to Earth (backbone)	2.2622	Alice \$50k at Earth, \$100k free at host Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.2625	Alice	Published geometry; own funding confirmed	Sends ORDER ×3 copies	Direct Ea → Ma, launches 2.2625, 2.2791, 2.2958; p(loss) 0.129 each, all lost 2.1e-03	2.5024 (first)	Alice \$50k at Earth, \$100k free at host Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.5024	Mars exchange	Its own available balances	Admits Alice EQ-buy; holds \$100k	—	—	Alice \$50k at Earth, \$100k held Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.5024	Mars exchange	Both holds in its own ledger	Fill 1,000 Ares @ \$100: Alice buys from Bob	—	local, atomic	Alice \$50k at Earth, 1,000 Ares at market Bob \$150k at Mars, 2,000 Ares home
2.5024	Mars exchange	Instruction or fill; its own ledger	Debits Alice 1,000 Ares; TRANSFER MA-X1 (automatic home delivery) to Earth	New session (SYN h 0.0006); DATA Ma → A h 2.5027, A → Ea h 2.7549	3.0674	Alice \$50k at Earth, 1,000 Ares in transit Bob \$150k at Mars, 2,000 Ares home
3.0674	Earth exchange	Authentic new transfer MA-X1	Imports once; credits Alice 1,000 Ares (spendable now)	RECEIVED to Mars (backbone)	3.6331	Alice \$50k at Earth, 1,000 Ares home Bob \$150k at Mars, 2,000 Ares home

S1d full trace: capped futures, rising path (Alice long, Cara short, Q = 2,000)

h	Actor	Knows	Does	Packet / transmission	Arrives	Financial state after
0.0003	Alice	Own Earth balance	Funds own Mars account: \$80k	Local access at Earth	0.0006	Alice \$150k at Earth Cara \$100k at Ceres
0.0003	Earth exchange	Instruction or fill; its own ledger	Debits Alice \$80k; TRANSFER EA-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ea → A h 1.1317, A → Ma h 1.4448	1.6966	Alice \$70k at Earth, \$80k in transit Cara \$100k at Ceres

h	Actor	Knows	Does	Packet / transmission	Arrives	Financial state after
0.0003	Cara	Own Ceres balance	Funds own Mars account: \$80k	Local access at Ceres	0.0006	Alice \$70k at Earth, \$80k in transit Cara \$100k at Ceres
0.0003	Ceres exchange	Instruction or fill; its own ledger	Debits Cara \$80k; TRANSFER CE-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ce → A h 0.9918, A → Ma h 1.2351	1.4868	Alice \$70k at Earth, \$80k in transit Cara \$20k at Ceres, \$80k in transit
1.4868	Mars exchange	Authentic new transfer CE-X1	Imports once; credits Cara \$80k (spendable now)	RECEIVED to Ceres (backbone)	1.9826	Alice \$70k at Earth, \$80k in transit Cara \$20k at Ceres, \$80k free at host
1.6966	Mars exchange	Authentic new transfer EA-X1	Imports once; credits Alice \$80k (spendable now)	RECEIVED to Earth (backbone)	2.2622	Alice \$70k at Earth, \$80k free at host Cara \$20k at Ceres, \$80k free at host
1.9829	Cara	Published geometry; own funding confirmed	Sends CINSTR ×2 copies	Direct Ce → Ma, launches 1.9829, 1.9996; p(loss) 0.087 each, all lost 7.5e-03	2.1410 (first)	Alice \$70k at Earth, \$80k free at host Cara \$20k at Ceres, \$80k free at host
2.1410	Mars exchange	Instruction + own funded balance	Holds Cara margin \$80k	—	—	Alice \$70k at Earth, \$80k free at host Cara \$20k at Ceres, \$80k held
2.1410	Mars exchange	Contract state	Sends REPORT	Direct Ma → Ce, launches 2.1410; p(loss) 0.087 each, all lost 8.7e-02	2.2990	Alice \$70k at Earth, \$80k free at host Cara \$20k at Ceres, \$80k held
2.2625	Alice	Published geometry; own funding confirmed	Sends CINSTR ×3 copies	Direct Ea → Ma, launches 2.2625, 2.2791, 2.2958; p(loss) 0.129 each, all lost 2.1e-03	2.5024 (first)	Alice \$70k at Earth, \$80k free at host Cara \$20k at Ceres, \$80k held
2.5024	Mars exchange	Instruction + own funded balance	Holds Alice margin \$80k	—	—	Alice \$70k at Earth, \$80k held Cara \$20k at Ceres, \$80k held
2.5024	Mars exchange	Both margins held	Opens F1: escrow \$160k; maturity h 302.5024, fixing h 326.5024	REPORT to remote parties	—	Alice \$70k at Earth, \$80k margin Cara \$20k at Ceres, \$80k margin
2.5024	Mars exchange	Contract state	Sends REPORT	Direct Ma → Ea, launches 2.5024; p(loss) 0.129 each, all lost 1.3e-01	2.7422	Alice \$70k at Earth, \$80k margin Cara \$20k at Ceres, \$80k margin
2.5190	Mars exchange	Contract state	Sends REPORT	Direct Ma → Ce, launches 2.5190; p(loss) 0.087 each, all lost 8.7e-02	2.6771	Alice \$70k at Earth, \$80k margin Cara \$20k at Ceres, \$80k margin

h	Actor	Knows	Does	Packet / transmission	Arrives	Financial state after
62.5024	Mars exchange	Signed observation from Bob	Records observation 1 = 110	Local access	—	Alice \$70k at Earth, \$80k margin Cara \$20k at Ceres, \$80k margin
122.5024	Mars exchange	Signed observation from Bob	Records observation 2 = 120	Local access	—	Alice \$70k at Earth, \$80k margin Cara \$20k at Ceres, \$80k margin
182.5024	Mars exchange	Signed observation from Bob	Records observation 3 = 130	Local access	—	Alice \$70k at Earth, \$80k margin Cara \$20k at Ceres, \$80k margin
242.5024	Mars exchange	Signed observation from Bob	Records observation 4 = 135	Local access	—	Alice \$70k at Earth, \$80k margin Cara \$20k at Ceres, \$80k margin
302.5024	Mars exchange	Signed observation from Bob	Records observation 5 = 125	Local access	—	Alice \$70k at Earth, \$80k margin Cara \$20k at Ceres, \$80k margin
326.5024	Mars exchange	Fixing cutoff; uses observation 5	Fixes at 125: long net \$50k; gross long \$130k, short \$30k	—	discharge + backed claims	Alice \$70k at Earth, \$130k free at host Cara \$20k at Ceres, \$30k free at host
326.5024	Mars exchange	Instruction or fill; its own ledger	Debits Alice \$130k; TRANSFER MA-X1 (payout) to Earth	New session (SYN h 326.5027); DATA Ma → A h 327.6171, A → Ea h 327.8825	328.1736	Alice \$70k at Earth, \$130k in transit Cara \$20k at Ceres, \$30k free at host
326.5024	Mars exchange	Instruction or fill; its own ledger	Debits Cara \$30k; TRANSFER MA-X2 (payout) to Ceres	New session (SYN h 326.5029); DATA Ma → A h 327.5228, A → Ce h 327.7882	328.0321	Alice \$70k at Earth, \$130k in transit Cara \$20k at Ceres, \$30k in transit
328.0321	Ceres exchange	Authentic new transfer MA-X2	Imports once; credits Cara \$30k (spendable now)	RECEIVED to Mars (backbone)	328.5422	Alice \$70k at Earth, \$130k in transit Cara \$50k at Ceres
328.1736	Earth exchange	Authentic new transfer MA-X1	Imports once; credits Alice \$130k (spendable now)	RECEIVED to Mars (backbone)	328.7309	Alice \$200k at Earth Cara \$50k at Ceres

S1e (falling) is step-for-step identical to S1d until the first observation, as the margin rule requires: same opening (2.5024 h), same packets. Observations 90, 80, 70, 65, 75; fixing at 75 gives long net \$-50,000, so the gross allocations swap: Alice \$30,000, Cara \$130,000. Final home cash Alice \$100,000 / Cara \$150,000. Payout spendable: Cara 328.032 h, Alice 328.174 h.

Key steps of the other scenarios (hours)

Steps
S1b funds at Ceres 1.670, receipt known 2.226, fill 2.608, Cara spendable 2.608, Belt home 3.164. Same structure as S1a, Earth–A–Ceres; no Mars involvement.
S1c funds at Earth 7.913, receipt at Uranus 10.551, fill 13.188, Fin spendable 13.188, Terra at Uranus 15.824. Eve sends 8 copies Uranus → Earth (p(loss) 0.78 each).
S1f As S1d with \$100k margin each: open 2.502, payouts spendable 328.03 / 328.17; Alice ends \$212,500, Cara \$37,500.

Steps

S1i Alice funds \$10k at Mars; batched direct packet carries BUY 40 @ \$100 (o1) and BUY 20 @ \$100 (o2): o1 fills 10 @ \$97 (price improvement \$30 refunded at once) and 5 @ \$100; o2 cancelled at +6 h releasing \$2,000; she sells 10 @ \$101 to Bob at +12 h; at +24 h her cancel of o1 races Bob's IOC sell: the earlier arrival fills 5 @ \$100, the cancel then releases only the remainder. Final sweep: cash and Ares home, done 51.06 h, 59 packets for 7 transactions.

E2. Probability and traffic

Direct service (exact). Each copy is lost independently with $p = 1 - e^{-(0.08 d)}$ at its own geometry; copies 60 s apart do not share a closure (we check the path is open for every copy) and incidents are excluded from these figures. $P(\text{all lost}) = \prod p_i$. A lost order is *not* a financial failure: the client's cash simply stays available at the host until the fallback copy arrives.

Run	Sender	Path	What	Copies	p each	P(all lost)	P(≥ 1 arrives)	If none arrives
S1a	Alice	Earth → Mars	ORDER	3	0.1289	2.14e-03	0.99786	resend same ID at +2(direct+bb)+6 h
S1c	Eve	Uranus → Earth	ORDER	8	0.7815	1.39e-01	0.86082	resend same ID at +2(direct+bb)+6 h
S1d	Cara	Ceres → Mars	CINSTR	2	0.0867	7.52e-03	0.99248	resend at +2·direct+6 h until deadline
S1d	Mars exchange	Mars → Ceres	REPORT	2	0.0867	7.52e-03	0.99248	none needed (status only)
S1d	Alice	Earth → Mars	CINSTR	3	0.1289	2.14e-03	0.99786	resend at +2·direct+6 h until deadline
S1d	Mars exchange	Mars → Earth	REPORT	1	0.1289	1.29e-01	0.87114	none needed (status only)
S1i	Alice	Earth → Mars	ORDER o1+o2	3	0.1289	2.14e-03	0.99786	resend same ID
S1i	Alice	Earth → Mars	CANCEL o2	3	0.1288	2.14e-03	0.99786	resend same ID
S1i	Alice	Earth → Mars	ORDER o3	3	0.1287	2.13e-03	0.99787	resend same ID
S1i	Alice	Earth → Mars	CANCEL o1	3	0.1284	2.12e-03	0.99788	resend same ID
S1i	Alice	Earth → Mars	WITHDRAW cash+ARES	3	0.1280	2.10e-03	0.99790	resend same ID

Backbone (approximate, per launch geometry). For each hop of the S1a records: p of the data launch, $p(\text{unconfirmed}) = 1 - (1 - p_{\text{data}})(1 - p_{\text{receipt}})$, hop abandonment after four launches $\approx p(\text{unconfirmed})^4$ (geometry held at the first launch; the true value differs in the 4th significant figure), and the chance the data never crosses, p_{data}^4 . An abandoned hop triggers endpoint retries and then application resubmission of the same transfer ID; the money stays in transit, owned by the client, until the import. No financial state changes.

Record	Hop	launch h	d (AU)	p launch	p unconfirmed	P(abandon)	P(never crosses)
FU\$ \$ Earth → Mars	Earth → A	1.132	2.255	0.0441	0.0863	5.5e-05	3.8e-06
FU\$ \$ Earth → Mars	A → Mars	1.445	1.816	0.0357	0.0701	2.4e-05	1.6e-06
FU\$ \$ Earth → Mars	Mars → A	1.697	1.816	0.0357	0.0701	2.4e-05	1.6e-06
FU\$ \$ Earth → Mars	A → Earth	1.950	2.255	0.0441	0.0862	5.5e-05	3.8e-06
AUTO_FILL ARES Mars → Earth	Mars → A	2.503	1.816	0.0357	0.0701	2.4e-05	1.6e-06
AUTO_FILL ARES Mars → Earth	A → Earth	2.755	2.254	0.0441	0.0862	5.5e-05	3.8e-06
AUTO_FILL ARES Mars → Earth	Earth → A	3.068	2.254	0.0441	0.0862	5.5e-05	3.8e-06
AUTO_FILL ARES Mars → Earth	A → Mars	3.381	1.817	0.0357	0.0701	2.4e-05	1.6e-06

Lossy Monte Carlo (independent seeds, every launch drawn with the brief's law, our declared client policy). Mean packets are backbone + direct. Quota columns are the worst rolling-24 h use seen in any run: backbone originations by one exchange (limit 66) and direct packets by one principal (limit 12).

Run	Done	no-loss	median	p90	p99	max	mean pkts	bb quota	direct quota	resubmit runs	fallback runs
S1a	2,000/2,000	3.07	3.07	4.71	7.70	28.21	46.8 + 3.0	3	6	0	1
S1b	2,000/2,000	3.16	3.16	4.81	7.76	14.04	46.8 + 3.0	3	6	0	9
S1c	2,000/2,000	15.82	28.76	50.74	71.48	104.58	59.2 + 10.0	3	12	0	414
S1d	2,000/2,000	328.17	329.70	332.19	335.36	337.21	118.7 + 8.8	4	12	0	282
S1e	2,000/2,000	328.17	329.70	332.19	335.36	337.21	118.7 + 8.8	4	12	0	282

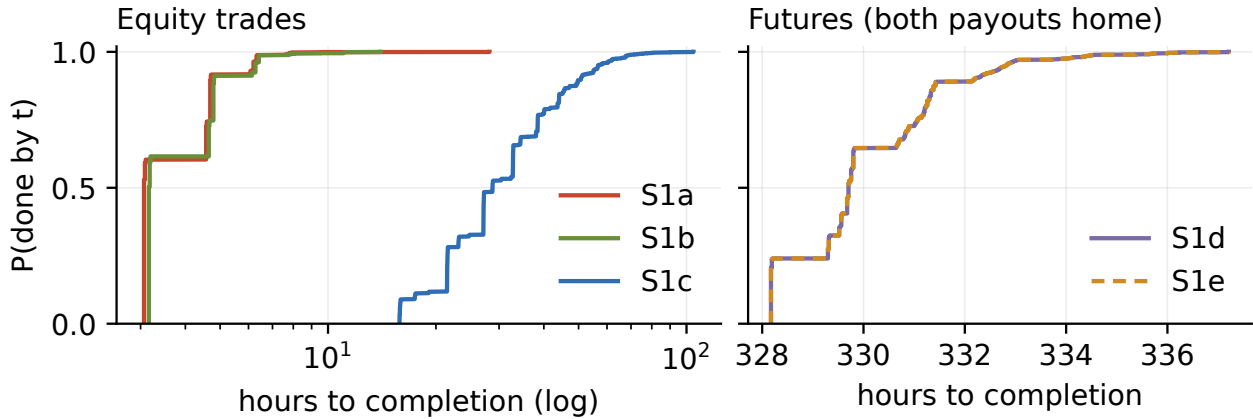


Figure A1. Probability of completion by time t under random loss (2,000 runs each).

Futures opening. The contract opened in 100.00% of lossy runs (median 3.63 h, p99 8.46 h), far inside the 168 h deadline. **Link capacity.** The longest FIFO queue on any directed link in any S1 run was 4 packets (limit 10,000); serialization never delayed a packet by more than a few seconds.

E3. Balances and capital

Conservation check. After every ledger event the simulator asserts, per asset: opening supply = Σ available + Σ order holds + Σ contract holds + Σ escrow + Σ unique exports not yet imported; every balance ≥ 0 ; every buy hold = remaining $\times$ limit; and, per registry, custody[s] = shares held at s + inbound + outbound in transit. No run ever failed a check. The futures figure in the design paper (Figure 3) plots balances over time; the table below gives every account's final position.

Run	Final balances (owner@location; anything not listed is zero)
S1a	Alice@Earth: \$50,000, 1,000 Ares; Fin@Earth: \$50,000, 1,000 Terra; Bob@Mars: \$150,000, 2,000 Ares; Cara@Ceres: \$100,000, 1,000 Belt; Eve@Uranus: \$50,000; Dax@Neptune: \$100,000
S1b	Alice@Earth: \$50,000, 1,000 Belt; Fin@Earth: \$50,000, 1,000 Terra; Bob@Mars: \$50,000, 3,000 Ares; Cara@Ceres: \$200,000; Eve@Uranus: \$50,000; Dax@Neptune: \$100,000
S1c	Alice@Earth: \$150,000; Fin@Earth: \$90,000, 600 Terra; Bob@Mars: \$50,000, 3,000 Ares; Cara@Ceres: \$100,000, 1,000 Belt; Eve@Uranus: \$10,000, 400 Terra; Dax@Neptune: \$100,000
S1d	Alice@Earth: \$200,000; Fin@Earth: \$50,000, 1,000 Terra; Bob@Mars: \$50,000, 3,000 Ares; Cara@Ceres: \$50,000, 1,000 Belt; Eve@Uranus: \$50,000; Dax@Neptune: \$100,000
S1e	Alice@Earth: \$100,000; Fin@Earth: \$50,000, 1,000 Terra; Bob@Mars: \$50,000, 3,000 Ares; Cara@Ceres: \$150,000, 1,000 Belt; Eve@Uranus: \$50,000; Dax@Neptune: \$100,000
S1g	Alice@Earth: \$150,000; Fin@Earth: \$50,000, 1,000 Terra; Bob@Mars: \$50,000, 3,000 Ares; Cara@Ceres: \$100,000, 1,000 Belt; Eve@Uranus: \$50,000; Dax@Neptune: \$100,000
S1i	Alice@Earth: \$149,040, 10 Ares; Fin@Earth: \$50,000, 1,000 Terra; Bob@Mars: \$50,960, 2,990 Ares; Cara@Ceres: \$100,000, 1,000 Belt; Eve@Uranus: \$50,000; Dax@Neptune: \$100,000

Ledger of the futures at its peak (S1d, h 2.502 to 326.502). Every dollar has one owner, one location and one use.

Asset lot	Owner	Location	Liability it backs	Encumbrance
\$80,000	Alice	Mars	Max loss of the long ($Q \times 40$)	Contract escrow, released only by fixing
\$80,000	Cara	Mars	Max loss of the short	Contract escrow
\$70,000	Alice	Earth	—	Free
\$20,000	Cara	Ceres	—	Free
\$50,000; 3,000 Ares	Bob	Mars	—	Free
\$100,000	Dax	Neptune	—	Free
\$50,000	Eve	Uranus	—	Free
\$50,000; 1,000 Terra	Fin	Earth	—	Free
1,000 Belt	Cara	Ceres	—	Free

Spendable, backed-claim and discharge moments for every settled position (no loss):

Position	Discharge (obligation ends)	Backed claim (funded export exists)	Spendable at home
S1a Alice shares	fill 2.502	fill 2.502	3.067
S1a Bob cash	fill 2.502	— (local)	2.502
S1d/e payouts	326.502	326.502	Cara 328.032, Alice 328.174

S2. Stress and recovery

How we chose the incident. We ran the futures scenario (both price paths) under every allowed incident on a grid: 72 h isolation of each settlement starting every hour, 6 h forced loss at each of the 11 nodes every 0.5 h, and an endpoint reset of each exchange every 0.25 h, all starting between h 0 and 420: 56,280 runs. Damage = extra hours until both payouts are spendable ($+10^6$ if the contract never opens). The worst start time for each kind and node:

Incident	Node	Start h	Opened h	Done h	Damage h
reset	Mars	0.75	128.13	453.79	+125.6
reset	Earth	327.25	2.50	453.73	+125.6
reset	Ceres	327.25	2.50	453.16	+125.0
isolation	Earth	2	80.23	405.89	+77.7
isolation	Mars	2	80.23	405.89	+77.7
isolation	Ceres	310	2.50	403.11	+74.9
forced loss	Earth	1.5	27.63	353.30	+25.1
forced loss	Mars	1.5	27.63	353.30	+25.1

Chosen: endpoint reset of the Mars exchange at h 0.75, the worst in both paths. It is not the obvious "isolate the host at payout time" (which costs about 75 h) but a short, silent event. At h 0.75 Mars has just answered Alice's and Cara's handshakes; the reset wipes its session state, but its SYN-ACKs are already in flight. Earth and Ceres therefore believe the sessions are up and send their funding transfers into sessions Mars no longer knows: Mars ignores them, as the brief requires. The senders learn nothing until four endpoint attempts of $R_e \approx 25$ h have expired, then wait the 24 h application backoff, then resubmit. Courier log (rising): h 0.00 Earth session open $\rightarrow$ Mars; h 0.00 Ceres session open $\rightarrow$ Mars; h 100.97 Ceres delivery unknown $\rightarrow$ Mars; h 100.97 Ceres app resubmit scheduled $\rightarrow$ Mars; h 101.64 Earth delivery unknown $\rightarrow$ Mars; h 101.64 Earth app resubmit scheduled $\rightarrow$ Mars; h 124.97 Ceres app resubmit $\rightarrow$ Mars; h 124.97 Ceres session open $\rightarrow$ Mars; h 125.64 Earth app resubmit $\rightarrow$ Mars.

State of every obligation during the incident. Alice's and Cara's \$80,000 are debited at home and "in transit", owned by them, unspendable anywhere, for about 125 h: no contract exists, so nobody owes anyone anything, and neither margin can be lost. Bob's shares and all other accounts are untouched. Local service at Mars never stopped: the reset only lost transport state. **Recovery** (financial service resumed, not just a packet): the transfers import at Mars, both margins are held, and the contract opens at h 128.13, inside the 168 h deadline, so the trade still happens with the full 300 h term and both payouts arrive home.

Path	Variant	Opened h	Done h	Packets	Originations
rising	no incident	2.50	328.17	120	12
rising	incident (baseline design)	128.13	453.79	176	16
rising	incident, maintenance removed	128.13	453.79	176	16
rising	no incident, maintenance removed	2.50	328.17	120	12
rising	incident + alternative A1: restart notice	3.82	329.49	168	18
rising	incident + sensitivity: no backoff	104.13	429.80	176	16
falling	no incident	2.50	328.17	120	12
falling	incident (baseline design)	128.13	453.79	176	16
falling	incident, maintenance removed	128.13	453.79	176	16
falling	no incident, maintenance removed	2.50	328.17	120	12
falling	incident + alternative A1: restart notice	3.82	329.49	168	18
falling	incident + sensitivity: no backoff	104.13	429.80	176	16

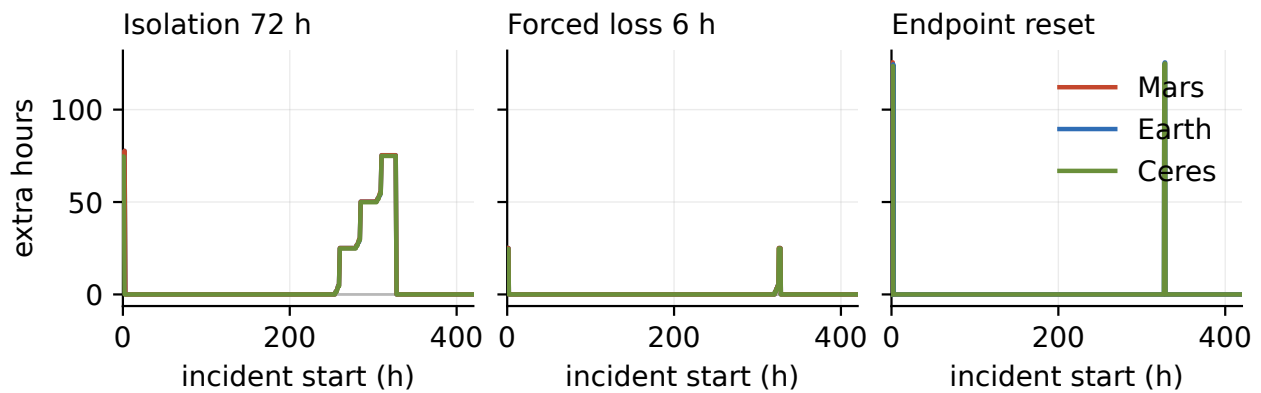


Figure A2. Extra hours to completion (rising path; capped at 200) by incident start. Grey: other nodes. Isolation and forced loss matter only around opening (h 0–3) and the payout window (h 255–330); a reset matters only when it lands between a handshake and the first data.

Maintenance comparison. Removing the two scheduled maintenance windows changes nothing for this incident: neither B–Neptune [2, 26] nor B–Ceres [240, 264] lies on the Earth–A–Mars or Ceres–A–Mars routes the runs use. Natural geometry alone sets the timing. **Alternative A1 (one design change, same funding and guarantees):** after a reset an exchange sends a RESTART notice (one 64-byte backbone record, counted against its quota) to every peer it has had a session with in the last 7 days; a peer receiving it abandons its old sessions with that exchange and resubmits unacknowledged records immediately. The contract now opens at h 3.82 instead of 128.13, with fewer packets. Removing the backoff alone (sensitivity) saves only the 24 h. We recommend adopting A1. Lossy Monte Carlo with the incident (1,000 runs each):

Variant	Done	median h	p90 h	p99 h	P(open)	mean pkts
incident	1000/1000	455.33	457.32	460.53	100.0%	185
incident_restart_notice	1000/1000	331.07	333.84	335.97	100.0%	176

S3. Access at every settlement

Transaction types: **funding / withdrawal / payout** (backbone, to the host exchange and back) and **orders / contract instructions** (direct from the client to the host). Backbone cells: best route, one-way delay there/back in minutes, and availability = fraction of the 24 h from that hour in which every link can launch (1-min samples, maintenance applied). Direct cells: delay, 24 h availability and the copies our policy sends. Hosts: Mars (Ares + futures), Ceres (Belt), Earth (Terra).

From	Mars bb h0	Mars bb h300	Ceres bb h0	Earth bb h0	dir → Mars h0	dir → Mars h300	dir → Ceres	dir → Earth
Mercury	Me-A-Ma 42.4/42.4 100%	Me-A-Ma 42.0/42.0 100%	Me-A-Ce 41.8/41.8 100%	Me-A-Ea 46.0/46.0 100%	17m 95% ×3	17m 0% ×3	26m 100% ×4	11m 100% ×2
Venus	Ve-A-Ma 36.9/36.9 100%	Ve-A-Ma 35.9/35.9 100%	Ve-A-Ce 36.4/36.4 100%	Ve-A-Ea 40.6/40.6 100%	15m 100% ×3	14m 100% ×3	24m 100% ×3	3m 100% ×2
Earth	Ea-A-Ma 33.9/33.9 100%	Ea-A-Ma 33.4/33.4 100%	Ea-A-Ce 33.4/33.4 100%	local	14m 100% ×3	14m 100% ×3	23m 100% ×3	local
Mars	local	local	Ma-A-Ce 29.7/29.7 100%	Ma-A-Ea 33.9/33.9 100%	local	local	9m 100% ×2	14m 100% ×3
Ceres	Ce-A-Ma 29.7/29.7 100%	Ce-A-Ma 30.5/30.5 100%	local	Ce-A-Ea 33.4/33.4 100%	9m 100% ×2	9m 100% ×2	local	23m 100% ×3
Jupiter	Ju-B-Ma 39.9/39.9 100%	Ju-B-Ma 39.2/39.2 100%	Ju-B-Ce 41.3/41.3 100%	Ju-B-Ea 50.9/50.9 100%	37m 100% ×4	36m 100% ×4	34m 100% ×4	50m 100% ×5
Saturn	Sa-A-Ma 75.8/75.8 100%	Sa-A-Ma 77.1/77.1 100%	Sa-A-Ce 75.2/75.2 100%	Sa-A-Ea 79.4/79.4 100%	75m 100% ×7	76m 100% ×8	75m 100% ×7	70m 100% ×7
Uranus	Ur-A-Ma 154.6/154.6 100%	Ur-A-Ma 155.0/155.0 100%	Ur-A-Ce 154.1/154.1 100%	Ur-A-Ea 158.3/158.3 100%	150m 100% ×8	150m 100% ×8	141m 100% ×8	158m 100% ×8
Neptune	Ne-A-Ma 246.8/246.8 100%	Ne-A-Ma 248.3/248.3 100%	Ne-A-Ce 246.3/246.3 100%	Ne-A-Ea 250.5/250.5 100%	246m 100% ×8	248m 100% ×8	245m 100% ×8	240m 100% ×8

Ceres and Earth backbone cells at h 300 differ from h 0 by under 2 min (all routes 100% available); the only availability change in the whole table is Mercury's direct path to Mars, open 95% of the first day and shut for the whole of day 300 (closed h 22.5–357). **Relocation timelines.** Same client (a \$100k Ares purchase with home delivery from a comparable funded account) re-run from reset with its home moved to each settlement, ranked by no-loss completion. Best **Mars**: order sent 0.000, fill 0.000, Bob paid 0.000, shares home 0.000. Median **Mercury**: funds at Mars 2.119, receipt home 2.826, order sent 2.826, fill 3.103, Bob paid 3.103, shares home 3.810. Worst **Neptune**: funds at Mars 12.344, receipt home 16.459, order sent 16.460, fill 20.565, Bob paid 20.565, shares home 24.681.

Home	no-loss h	pkts	median	p90	p99	P≤24h	P≤72h	P≤168h	mean pkts
Mars	0.00	0	0.00	0.00	0.00	1.00	1.00	1.00	0
Ceres	2.64	46	2.64	4.14	7.11	1.00	1.00	1.00	48
Earth	3.07	47	3.07	4.69	6.32	1.00	1.00	1.00	50
Venus	3.33	47	3.35	6.34	8.28	1.00	1.00	1.00	50
Mercury	3.81	47	3.83	7.22	9.53	1.00	1.00	1.00	51
Jupiter	3.95	48	3.98	7.27	12.29	1.00	1.00	1.00	51
Saturn	7.57	51	10.59	15.15	21.63	0.99	1.00	1.00	58
Uranus	15.38	52	26.76	48.49	70.31	0.34	0.99	1.00	68
Neptune	24.68	52	74.43	127.05	201.84	0.00	0.47	0.97	96

Outer settlements in the futures (300 lossy runs each; Dax at Neptune or Eve at Uranus is long against Cara; Eve's position is $Q = 1,250$ to fit her \$50k):

Long party	no-loss open h	no-loss done h	P(open by 168 h)	median open h	median done h
Dax / baseline	20.57	356.99	99.3%	61.55	423.35
Eve / baseline	12.80	344.56	100.0%	24.12	363.06

E4. Long-horizon scan — claiming Tier 3

Scan. 200 Julian years from the epoch at a 1 h step (1,753,201 epochs) for all 38 directed backbone links and all 72 directed direct paths, moving-receiver clearance at each sample, maintenance not repeated. Runtime 43 min on 20 cores. **Shortest closure we could miss:** clearance is the Sun's distance to the segment sender(t_e)→receiver(t_a), which changes no faster than the fastest endpoint plus the arrival shift: $L \leq 0.00146$ AU/h (Mercury at perihelion, 58.98 km/s). A closure entirely between two samples would leave a sample below $0.10 + L = 0.10146$ AU. Every sample below that guard is re-examined on a 10 s grid and every open/closed change is bisected to 1 ms, so no closure of any length is missed. **Routes:** best-route delay from every settlement to each market host and back, at 6 h steps, sequential flights.

Refined boundary (Tier 2). Earth → Relay A, first closure: hourly bracket [24001200, 24004800] s, bisected on the exact light-time solver to [24004585.374355, 24004585.375214] s (width 0.858 ms, tolerance 1 ms). A packet emitted before the boundary flies on its own path and is unaffected; one emitted after it is never launched by our queue (the link waits), so no packet in flight is lost by a closure: the closure only delays the next launch, and the hop timer R_h is restarted at the actual emission.

Backbone: Relay A–B never closes (min clearance 2.00 AU). Gateway links: availability · number of closures in 200 y · longest closure. Routes: one-way minutes min–median–max over 200 y. Direct to Mars: availability · longest closure.

From	to A	to B	→ Mars min	→ Ceres min	→ Earth min	Direct → Mars
Mercury	90.95% · 788 · 267 h	90.94% · 789 · 267 h	30–42–86	20–43–91	35–43–84	89.7% · 14 d
Venus	94.62% · 283 · 345 h	94.62% · 283 · 345 h	27–43–87	19–43–91	33–44–85	93.6% · 24 d
Earth	95.69% · 158 · 493 h	95.69% · 158 · 493 h	25–43–87	17–43–92	local	94.7% · 53 d
Mars	96.88% · 64 · 1086 h	96.88% · 64 · 1084 h	local	12–44–94	25–43–87	local
Ceres	99.49% · 2 · 4565 h	99.69% · 2 · 3768 h	12–44–94	local	17–43–92	98.9% · 38 d
Jupiter	98.34% · 25 · 1243 h	98.36% · 25 · 1211 h	30–59–112	23–65–106	33–58–110	97.4% · 26 d
Saturn	98.99% · 35 · 747 h	98.95% · 36 · 749 h	62–95–148	53–99–144	67–92–145	97.8% · 20 d
Uranus	98.73% · 40 · 571 h	98.73% · 40 · 571 h	139–174–227	134–178–230	145–172–227	97.8% · 17 d
Neptune	99.03% · 41 · 528 h	99.03% · 41 · 529 h	235–263–315	227–268–309	240–261–312	98.0% · 17 d

Every settlement had a route to every host at every sample (connected fraction 1.000 for all 42 directed settlement–host pairs), as the geometric argument in the design paper predicts. **Poorest service:** the longest one-way route is Mars → Neptune at 315 min (h 993,510); the longest direct closure is Jupiter → Saturn at 70.8 days. Direct paths are the weak link: their closures are what our access check, deferral and option O2 exist for.

E5. Shifted epochs — claiming Tier 3

Each run starts at $t = k \times 365.25$ d with every orbit advanced from the epoch, balances reset to the opening book, maintenance and the S2 incident not replayed, and observation/maturity times shifted with the start. Cells: no-loss completion h (median of 300 lossy runs); "· O2" shows the result with option O2 where it differs. S1d and S1e give identical times at every epoch (the margin rule is symmetric; only the payout split differs), so S1e is listed in the JSON results only.

Epoch	t_0 (h)	Routes used (one-way min, 24 h availability)	S1a equity	S1d/S1e futures
+1 year	8,766	Ea–Ma Ea-B-Ma 47' 100%, direct 17' 100%; Ce–Ma Ce-B-Ma 32', direct 100%	4.22 (5.76)	329.87 (331.90)
+10 years	87,660	Ea–Ma Ea-B-Ma 42' 100%, direct 22' 0%; Ce–Ma Ce-B-Ma 19', direct 100%	350.75 (352.10) · O2 3.87	suspended (—) · O2 329.27
+100 years	876,600	Ea–Ma Ea-B-Ma 39' 100%, direct 18' 100%; Ce–Ma Ce-A-Ma 50', direct 100%	3.52 (3.53)	330.48 (332.36)
Mars conjunction	12,470	Ea–Ma Ea-A-Ma 50' 100%, direct 20' 0%; Ce–Ma Ce-B-Ma 42', direct 100%	deferred (—) · O2 4.55	suspended (—) · O2 330.23

Difficult epoch (Tier 2–3). We chose the first Mars solar conjunction after the epoch: the Earth→Mars direct path closes at h 12,446.0 and reopens at h 13,713 (53 days); we start 24 h after closure. This is harder than any backbone event because client instructions cannot use the backbone. **Baseline:** Alice's client sees from the published geometry that no long-enough direct window exists, so it never funds Mars (equity deferred; Bob's 30-day sell order expires first, so no trade happens); in the futures Alice suspends at h 0 and Cara, who did fund, gets her margin back at the 168 h deadline and withdraws it home. **Funded consequences:** nobody loses anything; Cara's \$80,000 is locked for 168 h and returned by h 170.22. Both price directions behave identically because suspension happens before any observation. **With option O2** the instructions travel Earth→Mercury→Mars and both price runs complete normally.